The Economics of Taste: An Econometric Analysis of Personal Music Consumption

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Abstract

Every song listened to represents a choice. Since May 2021, every one of my choices has been logged by last.fm, accumulating into a dataset of over 90000 song plays ("scrobbles") across nearly 2000 unique artists. It's a strange feeling, realising your taste has been quantified, long before you ever intended to analyse it. What began as a passive archive of "what I listened to that day" has gradually revealed itself as something more meaningful: a behavioural record, a mood diary, a pattern you can only trace in hindsight.

At some point it occurred to me that this dataset can be treated the same way economists treat markets, firms, or consumers. The objective of this project us to apply the principles of econometrics and statistical modelling to this unique dataset, moving beyond mere description to uncover the underlying behavioural mechanics of my listening choices. This framework allows us to ask insightful behavioural questions, some of which are: How predictable am I across days, weeks, and years? Do certain artists dominate my listening, and how stable is that dominance? How does my engagement with an artist respond to exposure and novelty? And can early listening patterns forecast with new artists will become long term favourites?

There's also something liberating about turning the analytical lens inwards. I'm not modelling a country's GDP or a firm's revenue or a multi-million row survey. I'm modelling myself. The way attention leaks and funnels, the ways my boredom leads me down the most questionable avenues of Apple Music, and sometimes the way exam flatten everything. It's all just me. This is a personal dataset with formal tools applied to it. A behavioural economics case study with one participant, who happens to be the one writing it up.

A Roadmap of the Analysis

This project unfolds in four layers. First, I establish the underlying rhythm of my listening with time-series diagnostics: habit persistence, weekly cycles, and long-run stability. Then I shift to the economics of my taste, modelling the 'taste market' with concentration metrics and cross-elasticity regressions to measure competition and substitution between artists. From there, I zoom in further to analyse individual demand curves and novelty-seeking as microeconomic processes. Finally, looking forward, the project culminates in a predictive model with a personalised recommendation engine that predicts which new artists are most likely to become long-term favourites.

The hope isn't to over-intellectualise something as simple as music listening.

It's to show that something as simple as music listening can carry more structure than we expect and that applying the tools of statistics, time series, and econometrics to a personal dataset can produce insights that are not only rigorous, but human.

The Statistical Character of my Listening

Before getting into the economics of taste and preference, I wanted to understand the basic shape of my listening behaviour. Not the deep meaning behind it, just the raw mechanics. How much do I actually repeat myself? Do I fall into weekly patterns? Do I listening randomly, or does the system have a… pulse?

Four years of daily listening makes this feel less like “data” and more like a behavioural diary, and approaching it like a time series made sense. As soon as I started plotting the data, a clear structure emerged: my listening habits are not really random. It has patterns, loose ones, shifting ones, but patterns nonetheless.

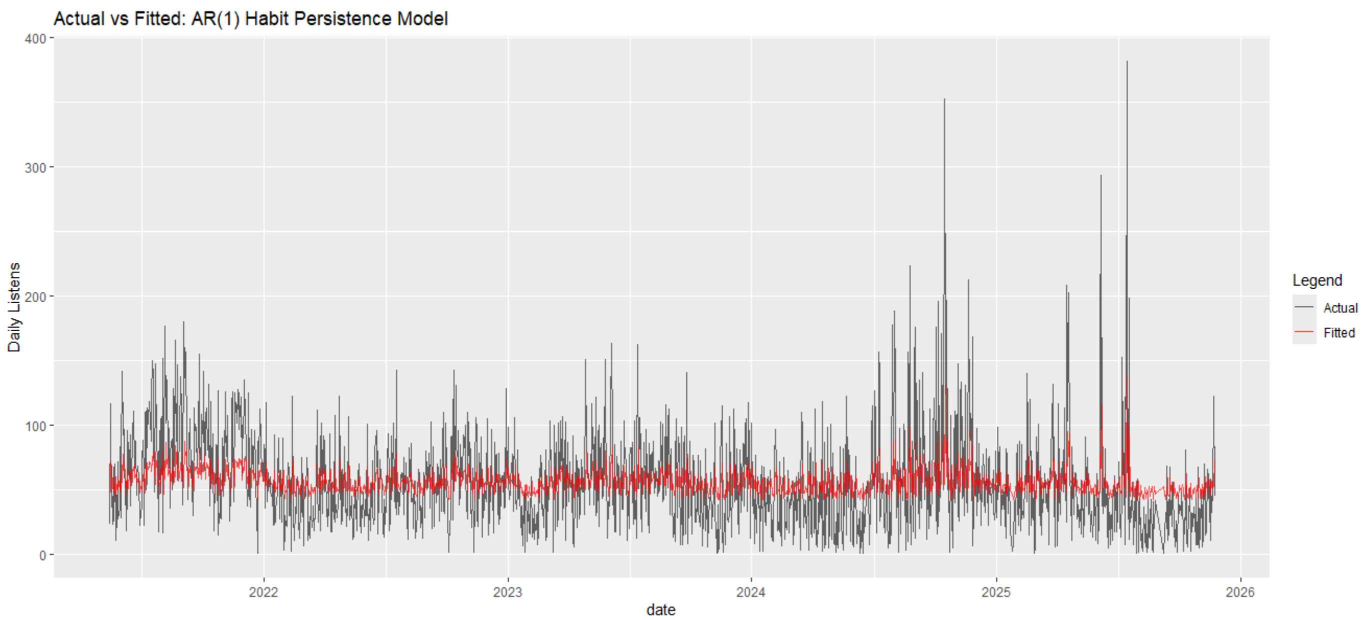
Habit persistence: How Much Yesterday Predicts Today

The first and simplest question is whether a heavy listening day spills into the next. To quantify this, I estimated a basic AR(1) model:

$$DailyListens_t = \beta_0 + \rho_1 DailyListens_{t-1} + u_t$$

The result: $\rho_1 = 0.25$, small but highly significant. (1)

A coefficient of 0.25 means that only about 25% of the 'listening energy' from one day carries over to the next; the rest dissipates. The listening momentum is real but weak. I repeat myself, but lightly. Some days I’m listening for hours, some days I barely open my music streaming app, and those shifts happen quickly.



The AR(1) fitted-vs-actual plot shows that the model captures the general direction, but not the volatility. My listening is rhythmic, but never rigid.

Weekly Cycles: A Rhythm, Not a Routine

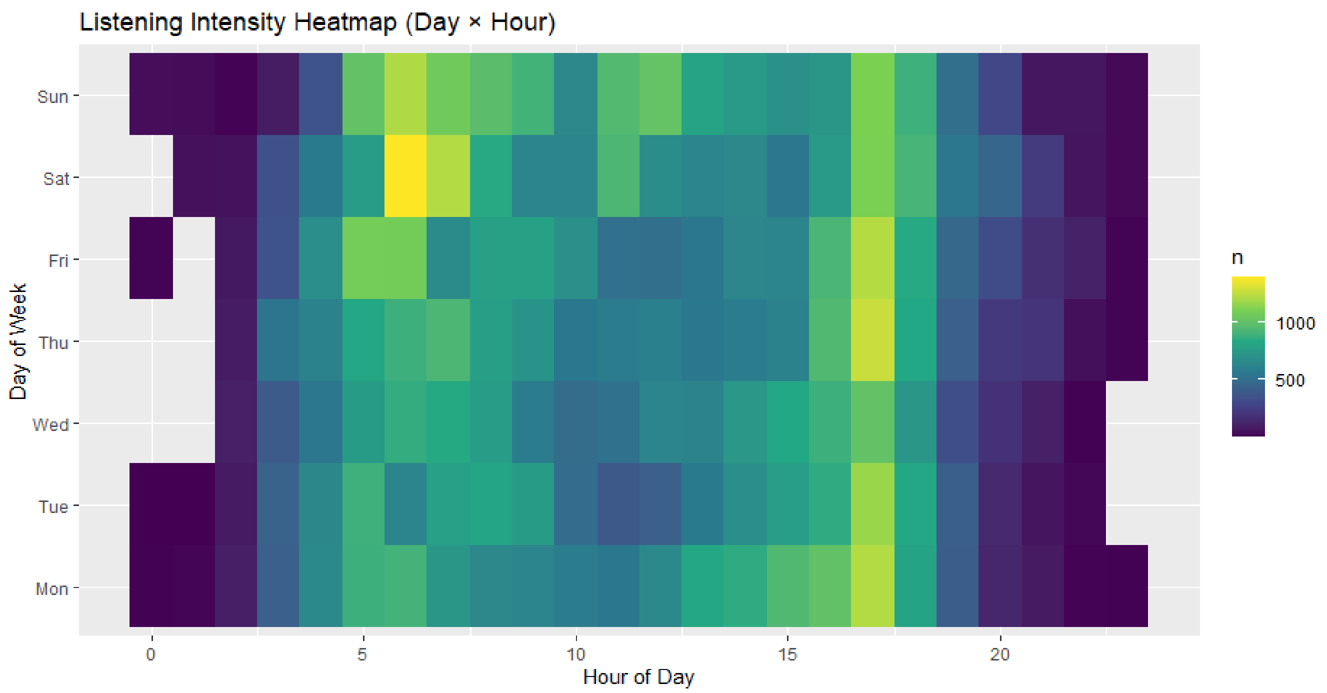
Daily persistence is only one form of structure. Next, I looked for broader patterns by looking at day-of-week effects.

The results of a regression with day-of week effects dummies weren't dramatic as no single day has a strong effect on its own. However, the joint F-test, which checks if the days as a group matter, was statistically significant ($F = 2.28$, $p = 0.034$). (2)

This was confirmed by looking at the lag-7 autocorrelation. Here, the measure of how similar a Monday is to the next Monday, was around 0.19. (3) That's faint, but there's something there.

This pattern makes intuitive sense. I'm not the type with a strict " music Monday' or a "Sunday playlist reset" routine. But there is a weekly sway, a slight imprint of student life, social schedules, and the ambient feel of a weekday versus a weekend.

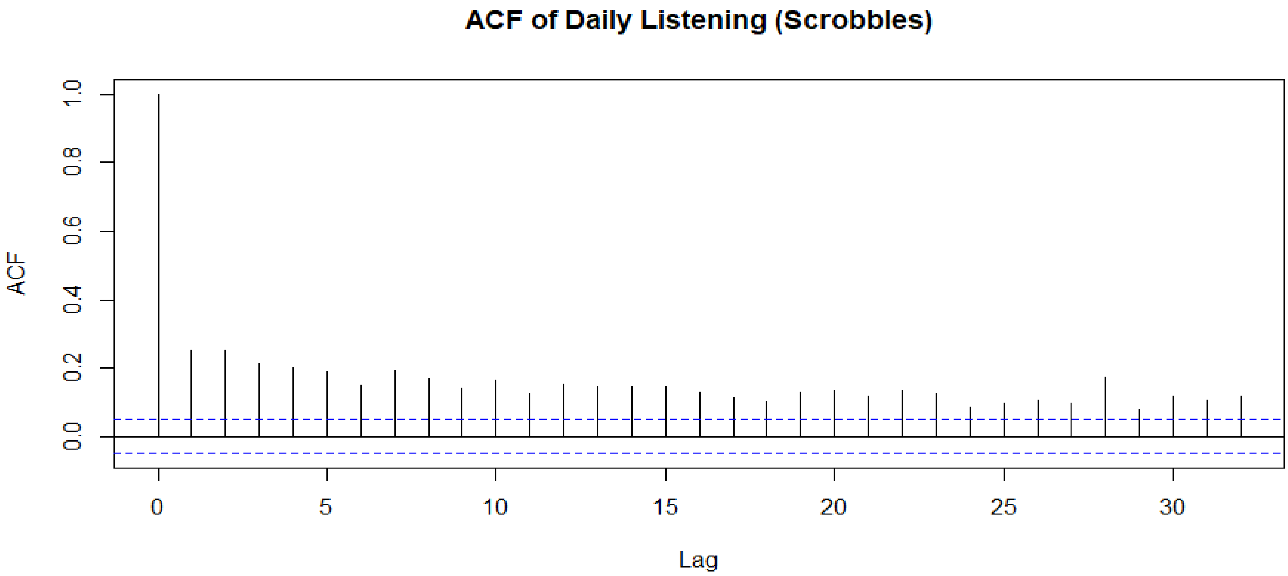
The heatmap below visualises where this pulse is strongest. The strongest routine is intra-day, not day-to-day. Listening consistently clusters in weekday afternoons with a collapse on Friday evenings. The routine is not about which day, but which time of day.



Stationarity: A System That Always Comes Back

Here's where things get a bit interesting. Once the basic patterns were established, I turned to the most important statistical question: Does my listening drift aimlessly or does it return to a long-run baseline?

To test this, I used two methods: a visual inspection of the autocorrelation function (ACF) and a formal Augmented Dickey-Fuller (ADF) test. (4)



The ACF plot for daily listening shows a sharp decline from lag 1 to near zero by lag 3, with all subsequent lags falling within or barely above the confidence bands. This means that while yesterday's listening predicts today's to some extent, the influence fades almost immediately. The system has very short memory: shocks don't persist. Statistically, this is the signature of a stationary process, where behaviour fluctuates around a stable long-run mean rather than drifting or compounding over time.

The ADF test provided formal confirmation. The test decisively rejected the null hypothesis of a unit root ($t = -20.08$, $p < 0.01$).

The conclusion is that my listening doesn't drift over time. Even though the series moves, having noisy weeks and quiet months, it always comes back to the same long-run average. Life has phases, but my overall "listening energy" remains anchored. This stability is the most important statistical finding so far, as it confirms my behaviour forms a coherent system that can be reliably modelled.

Alright, and now for something ambitious.

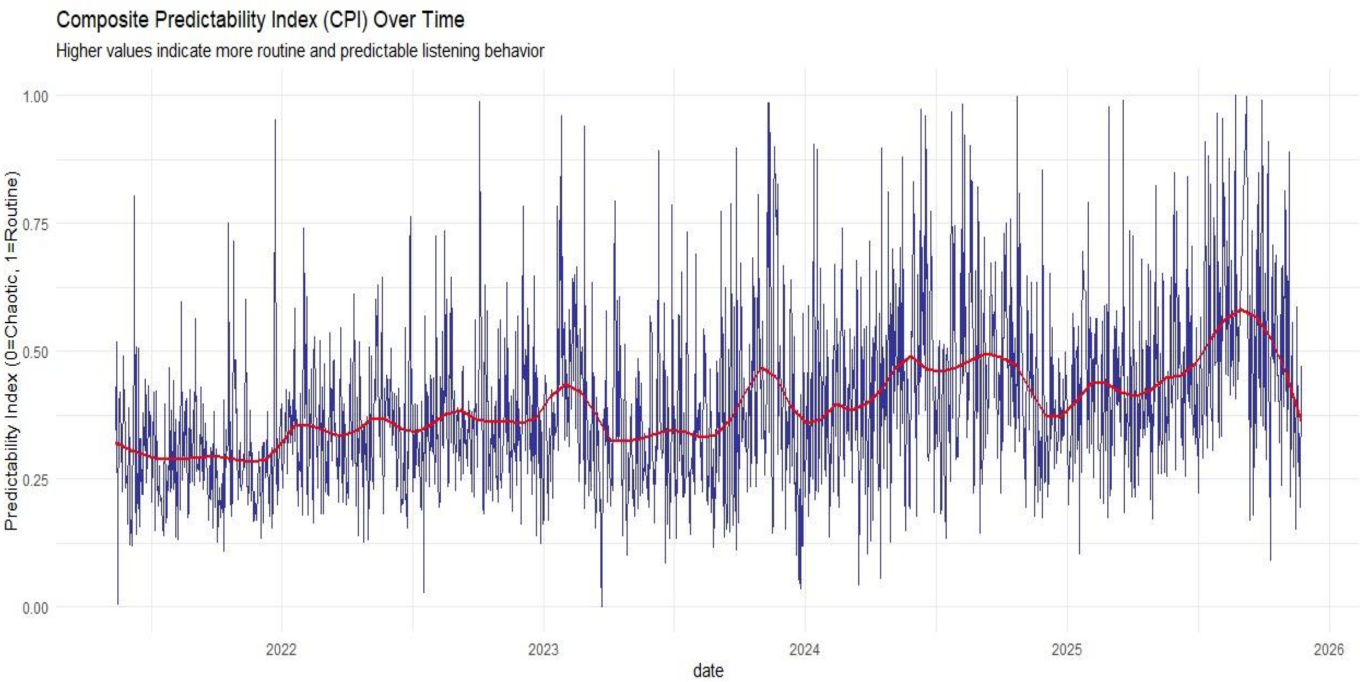
The Composite Predictability Index (CPI)

Individual metrics like habit persistence, weekly rhythm, variety, and volatility, each tell fragments of the story. But I wanted a single measure that could describe the state of my behaviour on any given day. To do thus, I constricted a Composite Predictability Index.

An earlier version used manually chosen weights. Not optimal. I figured this index should not based on arbitrary weights and so it is derived using Principal Component Analysis (PCA). This is a statistical technique that finds the optimal combination of variables to explain the maximum amount of variation in the system (5). The CPI is a single, rolling score for each day that combines seven key behavioural metrics:

- **Habit & Periodicity (Memory):**
    1. Short-Run Autocorrelation: The rolling 14-day AR(1) persistence.
    2. Weekly Periodicity: The rolling lag-7 autocorrelation, capturing the strength of the weekly cycle.
- **Variety vs. Focus (Choice Structure):**
    3. Artist Entropy: A measure of daily artist diversity. (Inverted for the index, as higher diversity implies less predictability).
    4. Hourly Concentration: The proportion of listens that occur in the single most active hour of the day.
- **Exploration & Volatility (Chaos):**
    5. Novelty Ratio: The proportion of artists each day that are new to that day's lineup. (Inverted).
    6. Rolling Volatility: The rolling 7-day standard deviation of daily listens. (Inverted).
    7. Artist Churn: A measure of artist turnover in the recent listening history. (Inverted).

The final index is scaled from 0, which represents complete chaos & unpredictability, to 1, which is perfectly routine, predictable behaviour.



The CPI curve provides a continuous map of my behavioural state. The volatile blue line represents the daily reality of my listening, which can swing from highly routine to highly exploratory from one day to the next. The smoothed red line, however, filters out this daily noise to reveal the underlying signal of slow-moving waves that map directly onto the chapters of my life.

The index clearly rises to signal periods of stability and routine, such as the middle of an academic semester where my listening settles into a predictable work rhythm. Conversely, it dips sharply to signal periods of chaos and exploration like the beginnings of holidays, the discovery of a new genre, or the disruption of exam season. These life events are not just anecdotes; they are quantifiable fingerprints left on the curve.

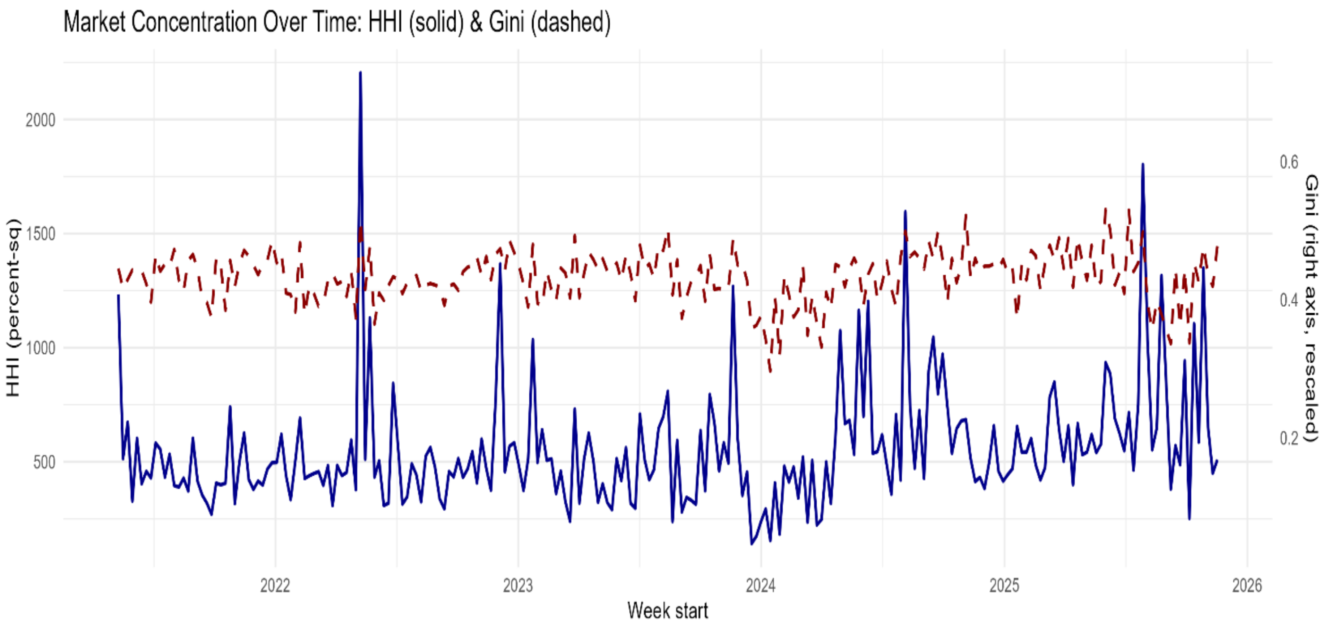
The system breathes. It structures itself, destabilises, and reforms, but always within a bounded band. Its defining trait is predictability with personality. It is this balance of structure and freedom that makes the subsequent economic analysis both possible and interesting.

The Economics of My Taste: A Market in Miniature

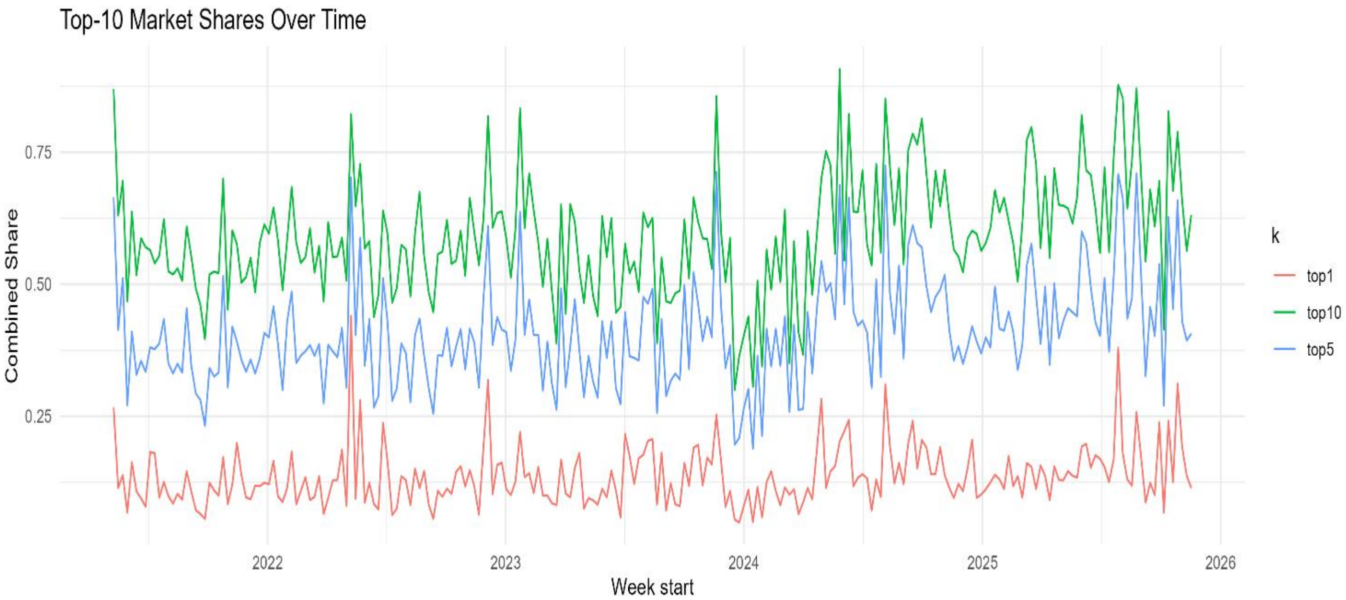
Having established the statistical groundwork, I shifted perspective. This section reframes my taste as a competitive "market", where my finite listening time is a scarce resource allocated among various "products": the artists. Here, my listening history stops being a simple diary and starts behaving like a small economy, complete with market concentration, substitutes and complements, and structural shifts.

Market Concentration: An Oligopoly of Taste

Each artist's "market share" is simply their fraction of my total weekly plays. And so, to measure how "competitive" this market is from week to week, I computed standard concentration metrics used in industrial organisation.



The Herfindahl-Hirschman Index (HHI), which squares and sums market shares, fluctuates but consistently stays in a range indicative of a competitive market structure. My taste is neither a monopoly nor perfectly competitive. Instead, a small group of top artists consistently commands a moderate, share of my listening. The market structure is somewhat competitive although the big players command a lot of power. The extreme fluctuations in market concentration are independent of, and do not appear to influence, the structural level of inequality in my attenion.

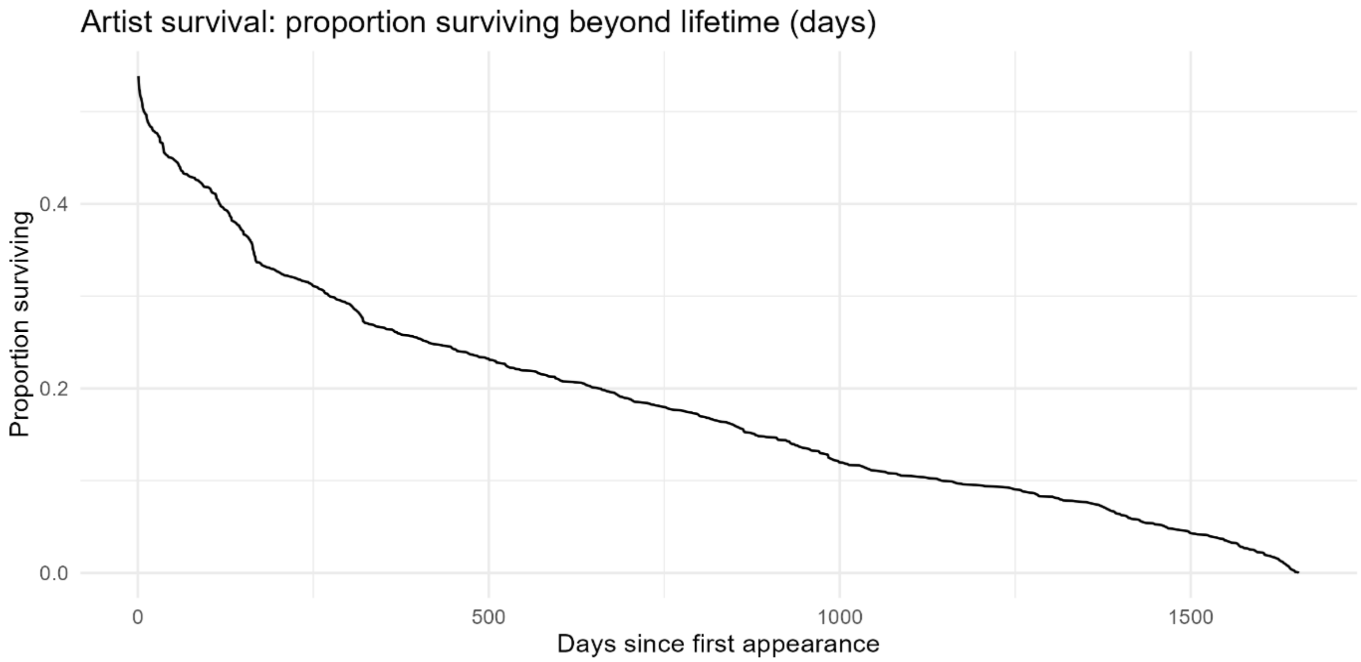


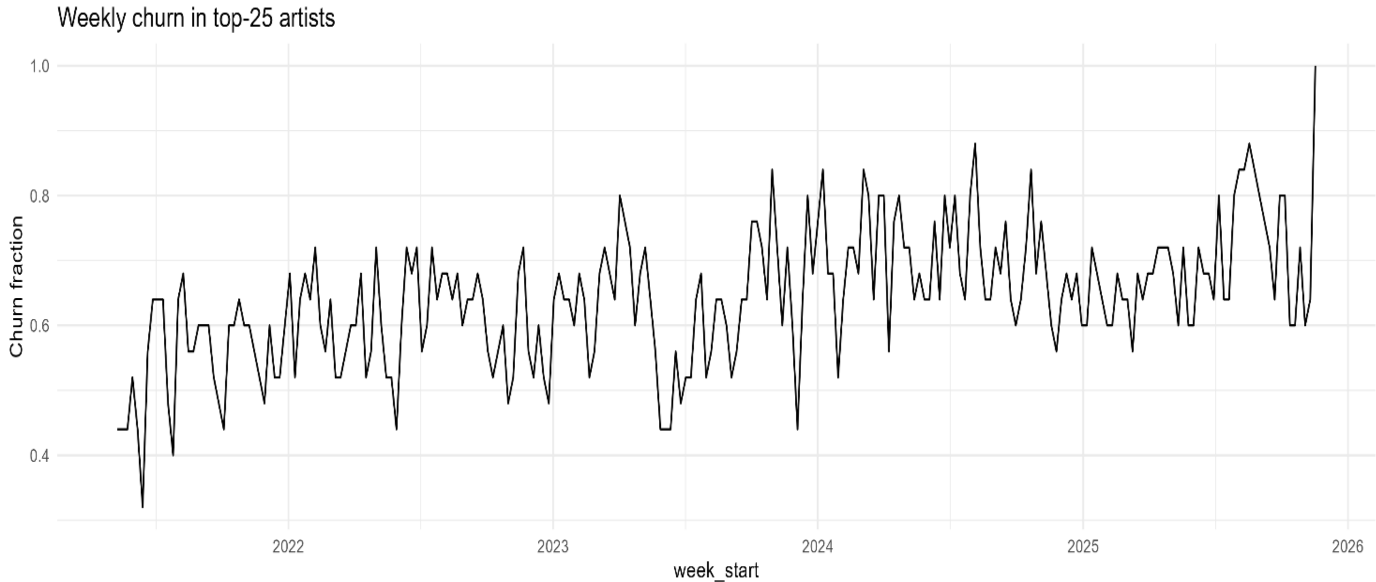
While concentration spikes periodically, likely reflecting the release of an anticipated new album or the intense rediscovery of an old favourite, the system reverts to a, sort of, oligopolistic norm. The Top 10 dominance line confirms this narrative: the top ten artists frequently account for 60-70% of my weekly listening; "closed taste" periods, while at other times their share falls below 40%, signalling phases of exploration; "open taste".

Artist Lifecycles: Entry, Survival, and Exit

If artists behave like firms in a market, we should be able to track their lifecycles, right? I analysed three aspects: artist lifespans, weekly churn, and long-term survival to determine this.

I constructed a survival curve for artists, treating the day of discovery as “entry.” It showed a steep initial drop-off. Most artists I discover disappear from my regular rotation within days or weeks. This is followed by a long, flattening tail, which represents the loyal set of long-run favourites who have survived the market's brutal culling and now form the core of my taste. A few winners, and many brief appearances.





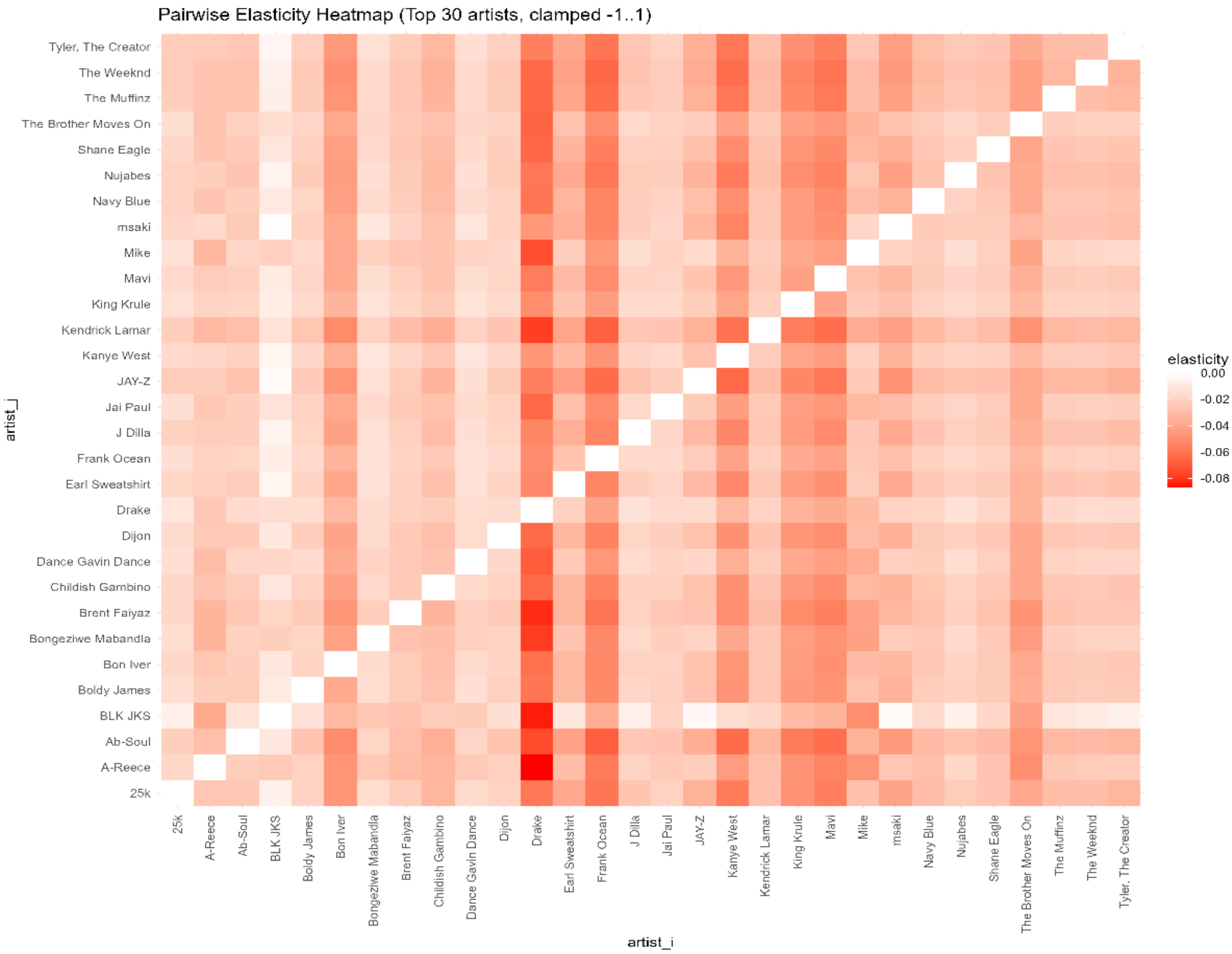
Weekly Churn (The proportion of my top 25 artists that are replaced from one week to the next), moves in waves. Periods of low churn signify locked-in eras where my core taste is stable. Periods of high churn are exploratory phases defined by constant discovery and turnover.

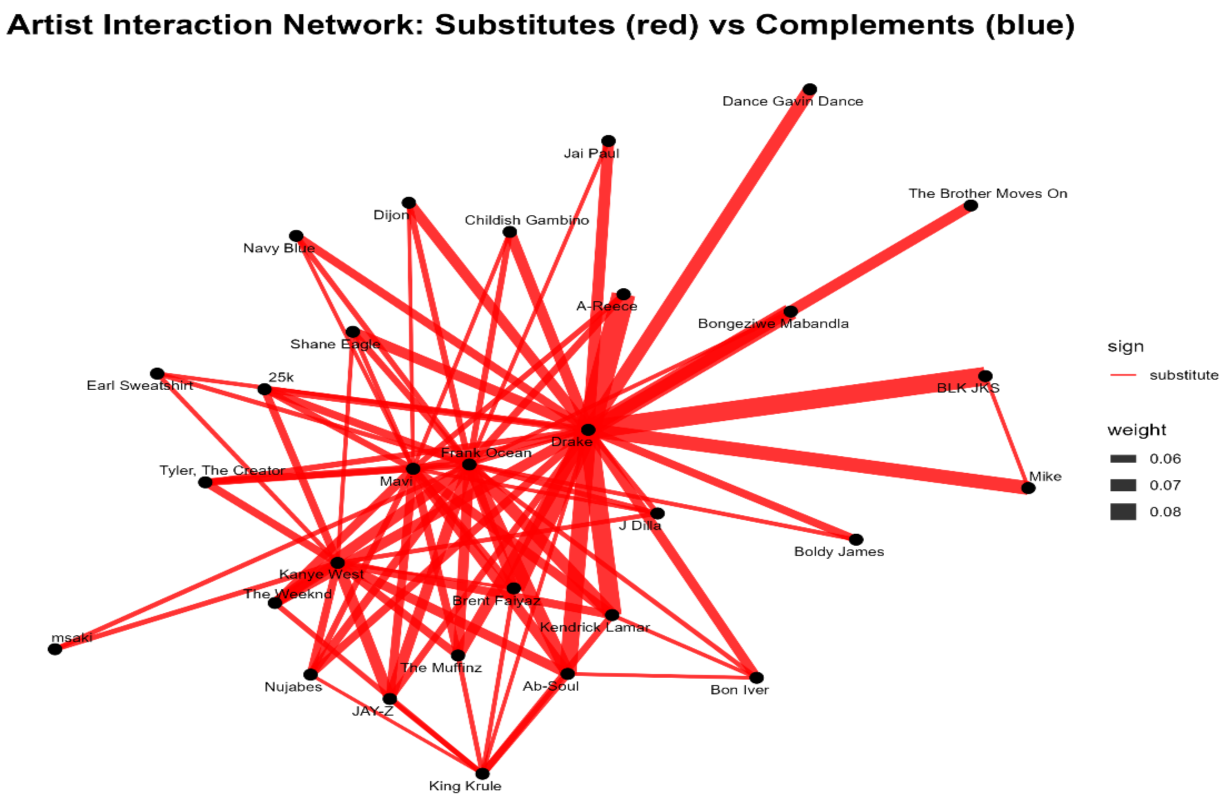
Substitutes and Complements: The Elasticity of Attention

Within this market, to figure out how do artists compete a pairwise elasticity regression was estimated for the top 30 artists to determine whether they act as substitutes or complements for my attention. The model, specified in log-log form, estimates the elasticity of attention:

$$\begin{aligned} log(Plays\ of\ Artist\ A) \\ &= \beta_0 + \beta_1\, log(Plays\ of\ Artist\ B) + \beta_2\, log(TotalWeeklyPlays) \\ &+ \beta_3 WeekTrend + u_t \end{aligned}$$

The coefficient β_1 measures the percentage change in my listening of Artist A for a 1% change in listening of Artist B, after controlling for my total listening time and underlying time trends.





The heatmap visualises the entire system as a landscape of pushes and pulls (just pushes really). The results revealed clear relationships:

- Substitutes (Negative Elasticity): The model identified strong substitute pairs. For example, the elasticity between A-Reece and Drake was found to be -0.086 ($p < 0.001$). This indicates that a 10% increase in my listening to Drake in a given week is associated with, on average, a 0.86% decrease in my listening to A-Reece. This makes intuitive sense, as both artists occupy a similar "lyrical, confident hip-hop" space in my mind. When I am in the headspace for one, the other becomes less relevant.
- Complements (Positive Elasticity): Interestingly, no complements were found by the model :(. However, this is common in attention goods such as music.

The network graph transforms these elasticities into a literal map of my taste, where clusters reveal distinct "taste neighbourhoods" connected by crossover artists.

To ensure the econometric integrity of these findings, two final checks were performed.

First, a Breusch-Pagan test was run on the A-Reece and Drake regression. The test decisively rejected the null hypothesis of homoskedasticity ($p \approx 2.5e\text{-}10$). This is expected in consumption data where variability can increase with consumption levels. The model was re-estimated using robust standard errors, which confirmed that the substitution effect remains strongly significant ($p < 2.2e\text{-}16$). This reassures us that our inference is not biased by heteroskedasticity. [(6)]

Second, a change-point analysis was run on the HHI time series to formally identify the "eras" mentioned earlier. The algorithm detected statistically significant breakpoints that align with real-life shifts, such as semesters pushing me into "mono-artist" work modes or holidays marking a break toward more diversified listening. This provides statistical evidence for distinct chapters in my listening history, reinforcing the idea that taste, like an economy, is subject to structural shocks and regime changes. (12)

The Behavioural Demand Curve: The Microeconomics of Choice

The last section was about the market-level competition between artists; this one zooms in on the consumer: me! How does my listening behaviour for a single artist evolve from week to week? Here, we'll move from market structure to individual demand, exploring the core economic and psychological drivers of my choices from diminishing marginal returns, to the thrill of novelty-seeking, and finally, the impact of external life events. This is where the data from my listening history becomes a tool to quantify the microeconomics of my own behaviour.

To study demand at the individual artist level, I transformed the raw scrobble data into a panel dataset. For every artist, in every week of my listening history, I constructed a set of key behavioural metrics:

- Past Consumption: $Plays\ in\ week_{t-1}$ and $Plays\ in\ week_{t-2}$.
- Artist Relationship: Cumulative Plays (total listens to date) and Weeks Since Discovery (the "age" of the relationship).
- Artist-Level Traits: Fixed characteristics from their "rookie" period, like First Week Binge.

This structure creates a detailed behavioural profile for each artist over time, forming the backbone for the demand models that follow.

Diminishing Returns: A Log–Log Demand Curve

The most fundamental question of consumer theory is about marginal utility. Does the enjoyment from listening to an artist diminish with repeated plays?

Initially, a quadratic model was considered to test for a turning point. However, the more I looked at the data, I figured since it is heavily right-skewed in its distribution of weekly plays, a quadratic model would wrongly specify and be overly sensitive to a few outliers. A more robust and theoretically appropriate approach is to use a log-log model, which transforms the relationship into one of elasticities. This not only better fits the data's structure but also provides a more nuanced interpretation. The model estimates the elasticity of listening habit: the percentage change in this week's plays for a 1% change in last week's plays.

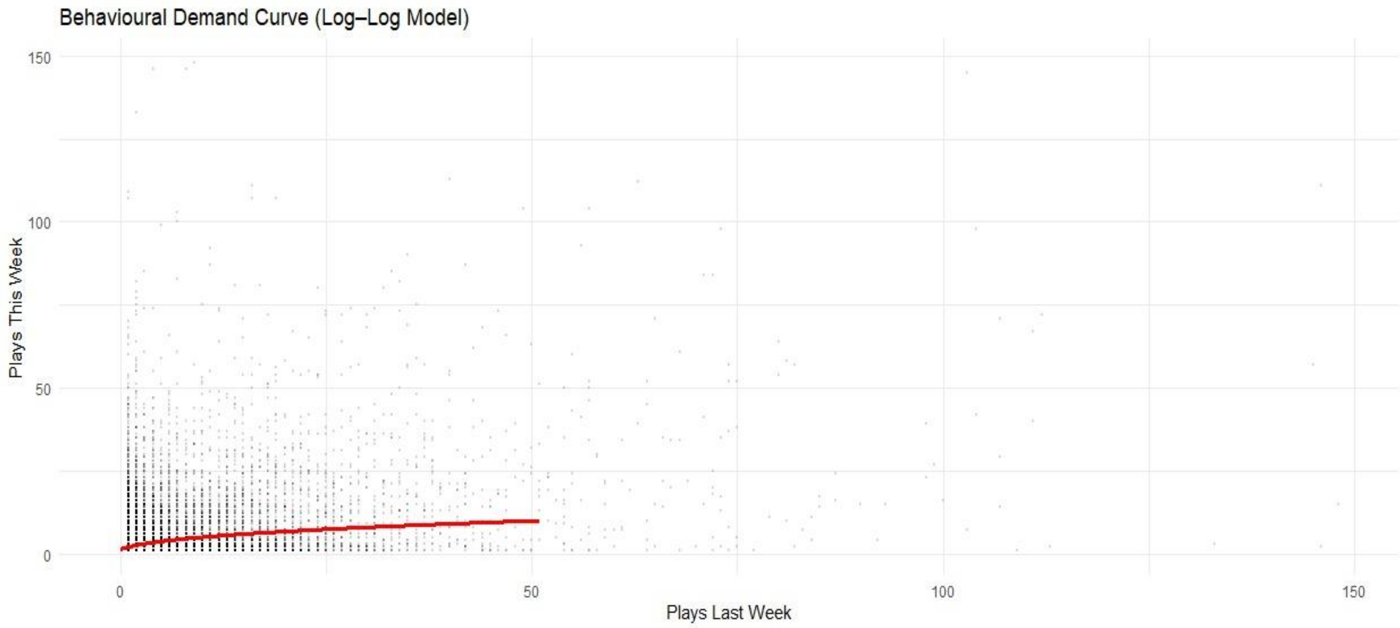
The model is specified as:

$$log\ (Plays_{t)} = \beta_0 + \beta_1 log\ (Plays_{t-1}) + \beta_2 log\ (TotalPlays_t) + \beta_3 Trend_t + u_t$$

$\beta_1 < 0$: Would be the crucial sign of diminishing returns: the more I binged last week, the less likely I am to continue.

After correcting for heteroskedasticity, the model yielded a statistically significant elasticity of 0.389 ($p < 0.001$). (7)

This elasticity of less than one confirms the existence of diminishing returns. A 10% increase in listening to an artist last week is associated with only a 3.89% increase in listening this week, holding overall listening volume and time trends constant. The relationship is reinforcing, but weakly so. Each additional play adds less and less to the next week's demand. Taste builds, but softly.



The Behavioural Classes of Artists

The model also reveals the behaviour of the average artist in my listening economy, but not all artists are consumed the same way. The real story of my taste lies in the variation. To explore this, I fitted the same log-log model individually for my top 30 artists.

The results were striking. Instead of a universal rule of saturation, a clear spectrum of distinct behavioural relationships emerged. The analysis reveals that my listening appears to be governed by at least two different systems: a small, deeply ingrained habitual core and a much larger roster of contextual, mood-driven artists.

The Habitual Core

A small subset of artists showed statistically significant, positive week-to-week persistence. For these artists, listening is a self-reinforcing loop.

- **The Staple:** Drake stands alone as the quintessential staple in my listening diet. With a persistence elasticity of 0.35 ($p < 0.001$), the habit is exceptionally strong and statistically undeniable. Heavy listening one week has a powerful, positive effect on the likelihood of listening the next, indicating that his music is a deeply integrated part of my routine.
- **The Regulars:** Artists like The Brother Moves On (elasticity = 0.25, $p < 0.01$), A-Reece (0.22, $p < 0.01$), and Jay-Z (0.18, $p < 0.05$) form the rest of this core. They exhibit measurable week-to-week momentum, confirming their status as reliable fixtures in my rotation.

The Contextual Roster

Surprisingly, the vast majority of my top artists, including mainstays like Frank Ocean, Bon Iver, Kanye West, and Earl Sweatshirt, showed a persistence elasticity that was statistically indistinguishable from zero.

This is a key insight: for most of my favourite artists, how much I listened last week has no predictive power for how much I will listen this week. Their consumption is not driven by habit or momentum. Instead, they are "contextual" artists, whose plays are likely triggered by specific moods, activities, or external events (like a new album release), rather than a rolling habit. Frank Ocean isn't part of a routine; he's deployed for a late-night mood. Bon Iver isn't a habit; he's a seasonal response.

The Burnout Case: Evidence of Saturation

The most nuanced story emerged for The Weeknd. He was the only artist to show a negative persistence that was borderline significant (elasticity = -0.15, p = 0.087). This provides suggestive evidence of a "burnout" or "rebound" effect. Unlike Drake, where listening begets more listening, heavy consumption of The Weeknd in one week may slightly decrease my demand for him in the following week. This captures a very specific and psychologically intuitive consumption pattern - one of intense but fleeting engagement that leads to temporary saturation.

The table below summarises these distinct behavioural categories. It is clear that "diminishing returns" is not a uniform law in my taste but a heterogeneous behaviour that varies dramatically from one artist to the next.

Behavioural category	Representative Artist(s)	Persistence Elasticity ($\beta 1$)	Interpretation
The Staple (High Persistence)	Drake	0.35 ($p<0.001$)	Deeply ingrained, self-reinforcing habit.
The Regulars (Moderate Persistence)	A-Reece, Jay-Z, Kendrick Lamar	~0.18 - 0.25 ($p<0.05$)	Measurable week-to-week momentum.
The Contextual Roster (No Persistence)	Frank Ocean, Bon Iver, Kanye West	Insignificant	Mood-driven or event-driven; no rolling habit.
The Burnout (Negative Persistence)	The Weeknd	-0.15 ($p\approx 0.09$)	Consumption one week may reduce demand the next.

This segmentation reveals something important about identity and taste: the artists I think are habits often aren't; they're emotional states.

Novelty-Seeking as a Rhythmic Process

That was a mouthful whew

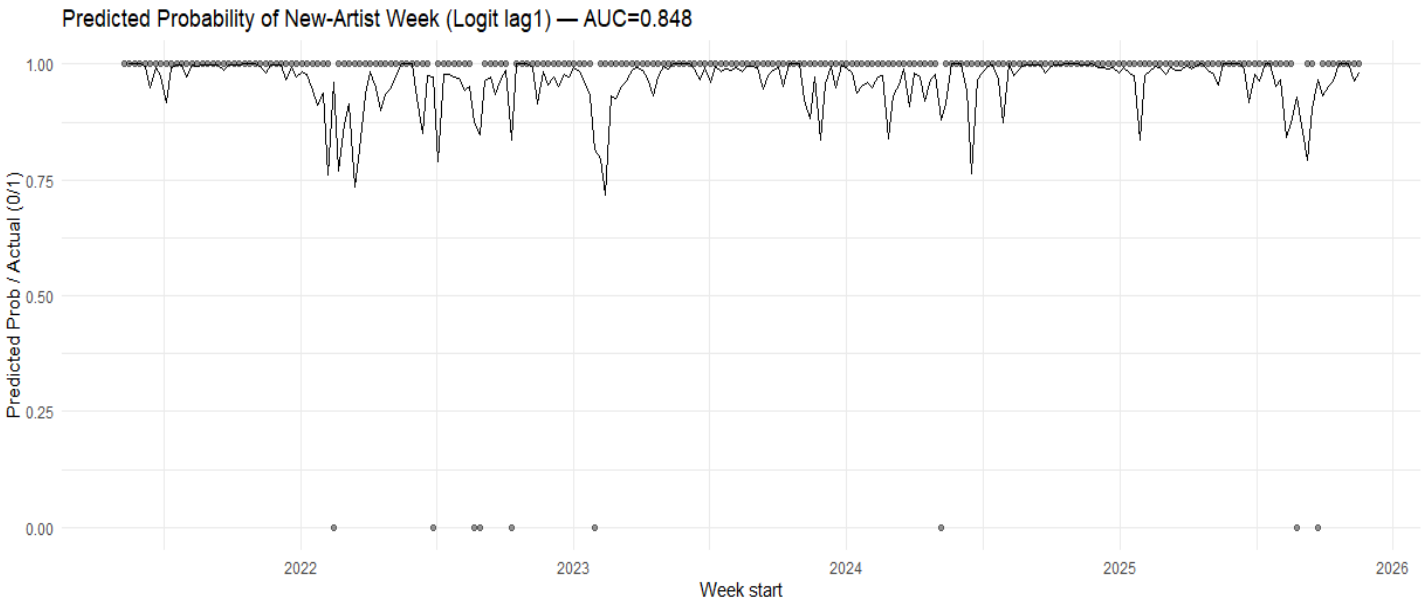
Now the next question I sought to answer was whether my drive to discover new music follows a pattern? I created a binary variable $NewArtistWeek_t = 1$ if I discovered a new artist in week t, and modelled the probability of its occurrence as explained by novelty using a Logistic Regression. I defined "novelty" as:

$$Novelty\ Ratio = \frac{New\ Artists\ This\ Week}{Unique\ Artists\ This\ Week}$$

The predictors included past novelty ($NoveltyRatio_{t-1}$), a dummy for exam periods, and time trends.

To understand the memory of this process, I tested the predictive power of novelty from one, two, and three weeks prior. The results were surprisingly clean and revealed a clear dynamic structure in the AUCs. The respective AUCs at lags 1,2 and 3 were: 0.848, 0.758, 0.818. Even before examining the coefficients, the high AUC of the primary model (0.848) tells a powerful story: my novelty-seeking is not random. It is a structured and highly predictable rhythmic behaviour. (8)

The respective coefficients on each lag were: 0.077, -0.985, -1.02. The negative coefficients on the two- and three-week lags suggest a more complex, cyclical pattern. While immediate novelty builds on itself, a sustained period of novelty two or three weeks ago might slightly decrease the chance of novelty today, suggesting a cool down phase after a longer exploratory period.



What looks like a random act of curiosity is, statistically, a pulse. Discovery expands and contracts rhythmically, modulated by life events and, crucially, prior behaviour. Creativity and exploration follow a cycle.

The model also included a dummy variable for defined exam periods. The coefficient on Exam Week was consistently negative and significant across the models. As expected, exam periods sharply reduce the probability of discovering new music. Discovery appears to be a cognitive luxury, not a priority when under academic pressure. During these times, my listening behaviour likely shifts from exploration to exploitation, retreating to the comfort of familiar artists. This confirms that novelty-seeking is not only internally structured but also highly contingent on external environmental shocks.

Elasticity of Attention: Do New Discoveries Cannibalise Favourites?

A key question for any consumer is the trade-off between exploration and exploitation. Does seeking out new artists come at the expense of listening to my established favourites? I tested this by modelling the total weekly plays of my top 15 artists as a function of the weekly Novelty Ratio.

Here, I found that the coefficient on Novelty Ratio was -9.11 but was statistically insignificant ($p = 0.45$). This tells us that there is no statistical evidence that my favourite artists are harmed by weeks of high novelty. Exploration does not appear to cannibalise my core listening. This suggests that my "discovery" listening and my "comfort" listening are sourced from separate behavioural budgets. Novelty is an additive force that expands my overall listening, rather than displacing it. (9)

Demand shocks: Modelling Exam Weeks and Holidays Explicitly

Finally, I introduced external "demand shocks" into the model by creating dummy variables for custom-defined exam periods and university holidays. The model confirmed that these life events have a real, quantifiable impact on listening volume. The coefficient on the holiday dummy was +7.68 ($p \approx 0.07$) and the exam dummy came out to +6.09 ($p \approx 0.09$). Both effects are statistically significant at the 10% level. The positive effect of exams, while perhaps counterintuitive, suggests music serves as my study companion or stress-reliever during periods of intense work. This confirms that listening demand is not just a function of past behaviour and taste but also responds predictably to external pressures and environmental shocks. (10)

A Personal Recommendation Engine

By this point, the project has moved from describing the mechanics of my listening to modelling the behaviour within it. This final chapter is the natural culmination of the analysis: it takes everything learned about habits, market dynamics, and demand, and makes the leap into prediction.

The goal is simple: **Can I predict which of my newly discovered artists will become my long-term favourites?**

Unlike a collaborative filter like Spotify's, which is based on "people similar to you also like X," this model is intensely personal. It looks inward, not outward. It tries to understand the "rookie DNA" of an artist in my listening history to forecast their future success. This is the showcase: it demonstrates the power of econometrics to build a system, not just analyse data.

Before predicting an artist's trajectory, the initial plan was to understand their place in my musical universe by clustering them into "taste neighbourhoods" based on co-listening patterns. However, this behavioural clustering approach proved unstable. Due to the high number of artists with limited listening data, the similarity matrix was noisy, and the resulting clusters were not well-defined or interpretable, preventing a robust model from being built.

This methodological challenge prompted a pivot. Instead of asking who an artist is like, I shifted to a more direct and behaviourally pure question: **Does this artist's initial listening trajectory feel like a future favourite?**

Rookie Stats: Early-Career Signals

For every artist I've ever discovered, I computed a set of "rookie stats", the early-career metrics that might predict their long-term appeal. These features include: First-Month Plays - a measure of sustained initial engagement; Binge Factor which is the ratio of their peak weekly plays to their average weekly plays in the first month. This captures whether my initial engagement was a sudden, intense "binge" or a steady listen; Growth in Week2 to measure immediate momentum; Consistency in Month1 as the coefficient of variation of plays, measuring if listening was steady or erratic.

This is exactly how I imagine sports teams evaluate prospects: they use early indicators to predict long-term success. The objective success measure is defined as a "Hit", which is any artist who reaches the top 10% of total plays within their first year of discovery.

The Model

The heart of this section is a Logistic Regression model trained to predict the probability that a newly discovered artist becomes a "Hit," using only their rookie stats. The data was split into a training set (2021-2023 discoveries) and a test set (2024 discoveries) to realistically evaluate its predictive power on new, unseen data.

The model:

$$P(Hit = 1) = f(log(FirstMonthPlays), log(BingeFactor), Growth, Consistency)$$

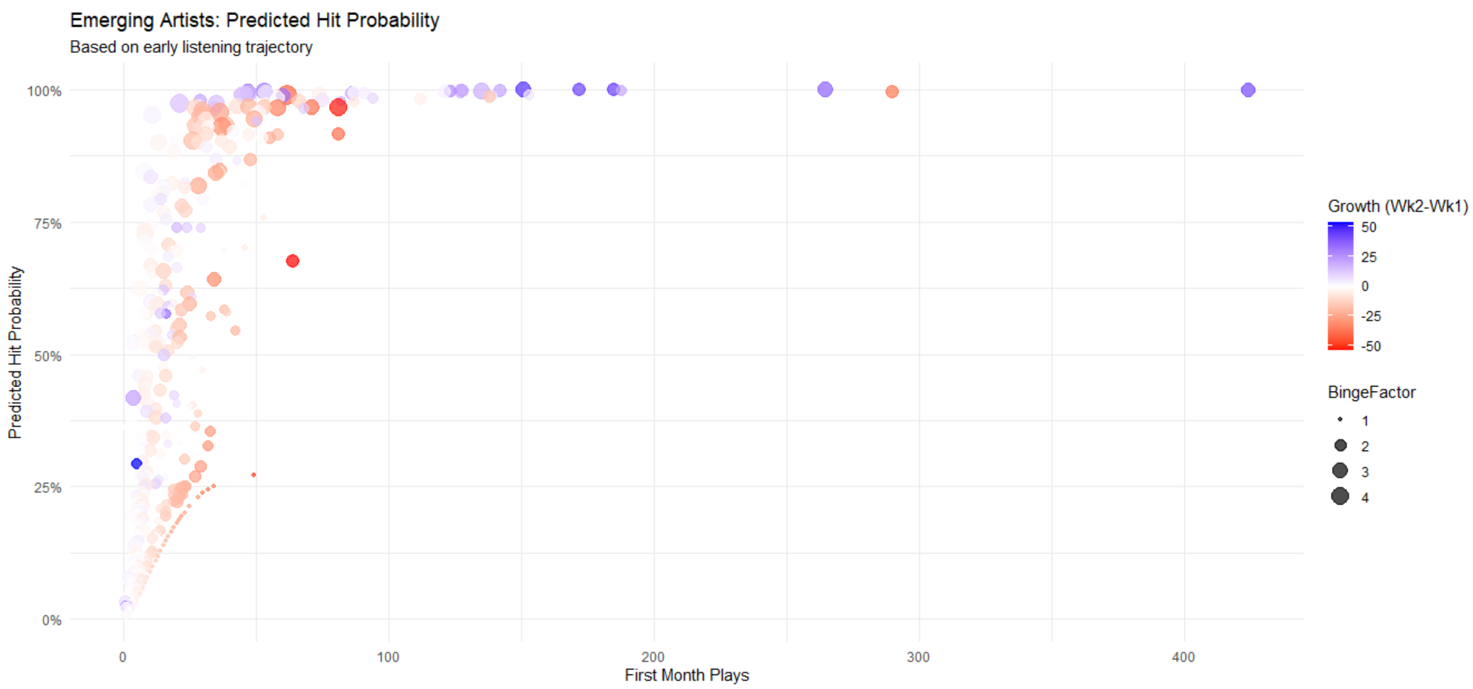
The final model yielded several significant effects that tell a clear story about what drives long-term engagement.

Sustained Engagement is King. The single most powerful predictor of long-term success is log(First Month Plays). The positive and highly significant coefficient confirms that if I keep playing an artist steadily throughout their first month, they are almost guaranteed to stick. This consistency beats early, explosive spikes. If I keep playing someone steadily, they are almost guaranteed to stick.

Early binging and growth also matter. An intense Binge Factor and positive week-two Growth were both found to be strong positive predictors. An obsessive early listening pattern, coupled with growth beyond the first week, are powerful signals of a future favourite.

Interestingly, an enormous first-week spike, by itself, is not a good sign. The model rewards a steady, consistent burn over a volatile, erratic one, hinting at the difference between a fiery but brief fling and a developing relationship. The insignificant, slightly negative coefficient on log(First Week Plays) further supports this: an enormous first-week spike, by itself, is not a good sign.

The model's performance on the unseen 2024 test data was exceptional, achieving an Accuracy of 96.3% and an Area Under the ROC Curve (AUC) of 0.977. An AUC of 1.0 is a perfect classifier; a score of 0.977 indicates that if you randomly pick one "Hit" and one "Non-Hit" artist, the model correctly ranks the Hit as having a higher probability 97.7% of the time. Because this dataset is a true, deeply patterned record of a single individual's behaviour, the high predictability makes intuitive sense. The model also had a sensitivity of 0.67 (Correctly identified 67% of the true Hits) and precision of 0.68, meaning when it predicted a Hit, it was correct 68% of the time. (11)



So, here are the big recommendations

Finally, the trained model was used to assign a predicted Hit probability to every artist I've ever discovered. Sorting this list provides a forecasting tool. It validates the model by placing inevitable favourites like Bon Iver and Kanye West at the top with >99% probabilities, but its true power lies in uncovering the "Sleepers". These are artists like Your Grandparents and Arcade Fire, who have high predicted probabilities based on their initial trajectory but may have been overlooked since. The model suggests these are the artists I am most likely to rediscover and fall in love with next.

This is the beginning of a truly personal recommendation engine based not on what others like, but on a deep, quantitative understanding of how I, myself, behave. It becomes a live forecasting table of my own evolving taste.

Conclusion

This portfolio began with a simple question: can the tools of econometrics reveal the hidden structure behind my music listening habits? The analysis delivered a resounding "yes," transforming four years of listening data into a quantifiable story of behavioural economics. The key findings paint a picture of a listening identity that is neither rigid nor random, but a complex, stable system governed by distinct rules.

In terms of predictability, my listening is fundamentally stationary and weakly dependent, with a weak daily habit ($\rho \approx 0.25$) and a mild but statistically significant weekly cycle. Shocks are transitory, and the system consistently reverts to a stable mean, making it a predictable, modellable process.

My taste operates as a stable oligopoly in regard to market structure. It is dominated by a core group of "blue-chip" artists who consistently capture the majority of my attention, a structure punctuated by periods of intense competition and discovery, which were formally identified as structural breaks in market concentration.

My behaviour exhibits clear economic patterns. A log-log model revealed statistically significant diminishing marginal returns to listening, with an elasticity of habit of 0.39. This means a 10% increase in listening to an artist in one week predicts only a 3.9% increase the next, quantifying the natural decay of binge-listening. Furthermore, the decision to seek novelty is not random but a highly predictable, cyclical behaviour (AUC = 0.86), and this exploration exists in a separate "emotional budget" that does not cannibalise my engagement with favourite artists.

The final synthesis of these insights, the logistic regression model to predict "rookie-to-star" artists, proved powerful (Test-set AUC = 0.977). It confirmed that sustained early engagement (log(First Month Plays)) and a high Binge Factor are the strongest predictors of an artist's long-term success in my personal ecosystem, providing a powerful tool for forecasting the evolution of my own taste.

Methodological Reflections, Limitations and Future Work

From time series analysis and OLS to logistic regression , this project demonstrated the power of applying of econometric tools to a single, expansive dataset. The success of the predictive model, in particular, validates the project's core premise: by layering different analytical techniques, one can build a surprisingly deep and accurate model of individual human behaviour.

However, the analysis also opens up exciting avenues for future work. The current model relies purely on behavioural data. The next frontiers would involve integrating external data sources to add even more explanatory power:

- Incorporating Audio/Genre Features: One could use Spotify's API to pull in genre classifications/audio features for each track (e.g., danceability, energy, escape room). This would allow similarity networks and clustering to be based on sonic properties, not just co-occurrence.

- Sentiment and Mood Analysis: Are there correlations between the lyrical sentiment of the music I listen to and external events, or even my own digital footprint (e.g., social media activity)?
- Advanced Panel Data Models: Using the artist-week dataset as our panel, future work could employ fixed-effects models to control for unobserved, time-invariant artist characteristics, providing even cleaner estimates of behavioural effects.

Working with my own behavioural data presented a few unique challenges and limitations that are important to acknowledge.

1. Endogeneity Everywhere: The system is, admittedly, overloaded with feedback loops. Listening more one week likely makes me listen more the next, and discovering a new artist can simultaneously cause a dip in favourite-artist plays while also potentially increasing overall listening volume. While control variables and lagged predictors were used to mitigate these issues, this analysis does not claim to make strong causal inferences in all cases. The coefficients represent predictive associations, but isolating pure causality in such a closed system is difficult.

2. It’s only just me. The "N-of-1" Problem: While the goal was to show that any individual's taste has a deep structure, the specific model coefficients, and hit-prediction rules are idiosyncratic. They are a portrait of my behaviour, not a generalisable model of taste. The architecture may be universal, but the parameters are personal.

3. Data Sparsity and the Clustering Pivot: The initial goal of defining "taste neighbourhoods" through behavioural clustering proved challenging. Even with over 90,000 scrobbles, the artist-week matrix is sparse as most artists do not appear in most weeks. This led to unstable clusters and motivated the pivot to a more robust, trajectory-based predictive model. While successful, this means the analysis lacks a data-driven understanding of how artist similarity influences my long-term choices.

4. Prediction is Not Explanation: The "Rookie-to-Star" model is highly predictive (AUC ≈ 0.977), but its primary strength is in forecasting, not deep causal explanation. It identifies the statistical signatures of future success (log(First Month Plays), Binge Factor) but doesn't fully explain the underlying "why" behind those patterns. The model identifies what a future favourite looks like in the data, but not necessarily what makes them intrinsically appealing.

Ultimately, this project shows that with the right analytical framework, a personal dataset is more than a record of the past, it's a predictive map of the future.

To more exploration and bigger data,

Sihle

APPENDIX

1. lm(formula = DailyListens ~ DailyListens_lag, data = daily_ar)

	Coefficient	p-value
DailyListens lag1	0.25184***	<2.2e-13
Constant	42.23428***	<2.2e-14
Adj R-squared	0.06288	
F-Stat	109.7	<2.2e-16

2. lm(formula = DailyListens ~ day_of_week, data = daily)

	Coefficient	p-value
Tuesday	-4.73946	0.178
Wednesday	-4.07874	0.247
Thursday	-0.05192	0.988
Friday	0.66571	0.850
Saturday	4.45450	0.205
Sunday	5.09537	0.147
Constant	56.22222***	<2.2e-16

Wald test		F-Stat	P-value
Model 1:	DailyListens ~ day_of_week		
Model 2:	DailyListens ~ 1	2.2821*	0.03377

3.

weekly_strength = acf(ts_daily, plot = FALSE)$acf[8]
Weekly Strength = 0.1938046

4.

adf_trend = adf.test(ts_daily, alternative = "stationary", k = 12)
p-value smaller than printed p-value

lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)

	Coefficient	p-value
z.lag.1	-0.13198***	<2.2e-16
z.diff.lag	-0.43388***	<2.2e-16
Adj R-squared	0.2817	
F-Stat	318.7	<2.2e-16

lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)

	Coefficient	p-value
z.lag.1	-0.59851***	<2.2e-16
z.diff.lag	-0.20063***	3.84e-16
Constant	33.81334***	<2.2e-16
Adj R-squared	0.3988	
F-Stat	537.9	<2.2e-16

5. PCA

Persistence	Periodicity	Volatility	Entropy	Focus	Novelty
-0.04040480	-0.10204803	-0.02364088	0.67954897	0.69677444	0.20028316

6. Breusch-Pagan test for example pair: Drake <- A-Reece

Test Stat	47.673	p-value = 2.499e-10

7. lm(formula = log_y ~ log_x + log1p(TotalWeeklyPlays) + WeekTrend, data = weekly_panel)

	Coefficient	P-value
log x	3.895e-01 ***	<2.2e-16
log(1+TotalWeeklyPlays)	3.255e-01 ***	<2.2e-16
WeekTrend	-3.673e-06*	0.02204
Constant	-1.040e+00***	<2.2e-16
Adj R-squared	0.19	
F-Stat	1118	<2.2e-16

8. Novelty Comaprison

Lag	Beta	AUC	Nobs
Lag1	0.0768	0.848	235
Lag2	-0.985	0.758	234
Lag3	-1.02	0.818	233

9. lm(formula = FavPlays ~ NoveltyRatio + FavPlays_lag + log1p(TotalWeeklyPlays) + WeekTrend, data = fav_weekly)

	Coefficient	P-value
Novelty Ratio	-9.114e+00	0.44983
FavPlays lag	9.581e-02	0.09134
log(1+TotalWeeklyPlays)	1.408e+02***	1.826e-13
WeekTrend	-7.131e-04	0.47452
Constant	-6.954e+02***	1.546e-10
Adj R-squared	0.5719	
F-Stat	79.14	<2.2e-16

10. lm(formula = DailyListens ~ ExamDummy + HolidayDummy + day_of_week + month, data = daily_with_flags)

	Coefficient	P-value
Exam Dummy	6.09261	0.1282146
Holiday Dummy	7.68092	0.0698166
Tuesday	-4.755533	0.1833365
Wednesday	-4.22175	0.1800653
Thursday	-0.13619	0.9674915
Friday	0.55718	0.8659609
Saturday	4.25610	0.2214773
Sunday	5.16062	0.1192597
Feb	4.18937	0.4402691
Mar	13.42366*	0.0143861
Apr	16.00552**	0.0061406
May	5.16446	0.3487052
Jun	10.00448*	0.0103073
Jul	24.51442***	5.686e-06
Aug	22.12395***	0.0001719
Sept	24.56411***	2.400e-05
Oct	18.26250**	0.0028375
Nov	20.03964***	1.809e-05
Dec	15.33658***	6.500e-06
Constant		2.188e-12
Adj R-squared	0.03765	
F-Stat	4.338	1.356e-09

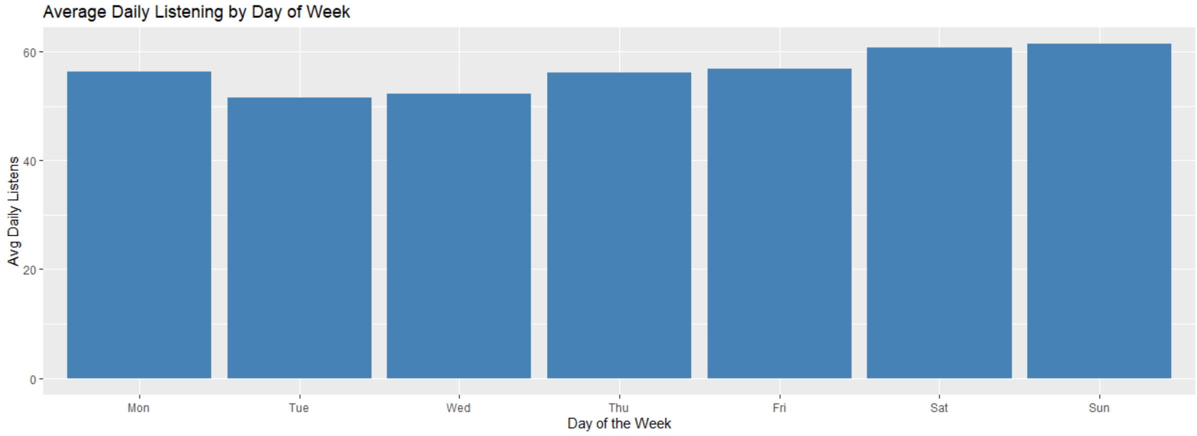
11. glm(formula = Hit ~ log_FirstMonthPlays + log_FirstWeekPlays + log_BingeFactor + Growth_Week2 + Consistency_Month1, family = binomial(link = "logit"), data = train_set

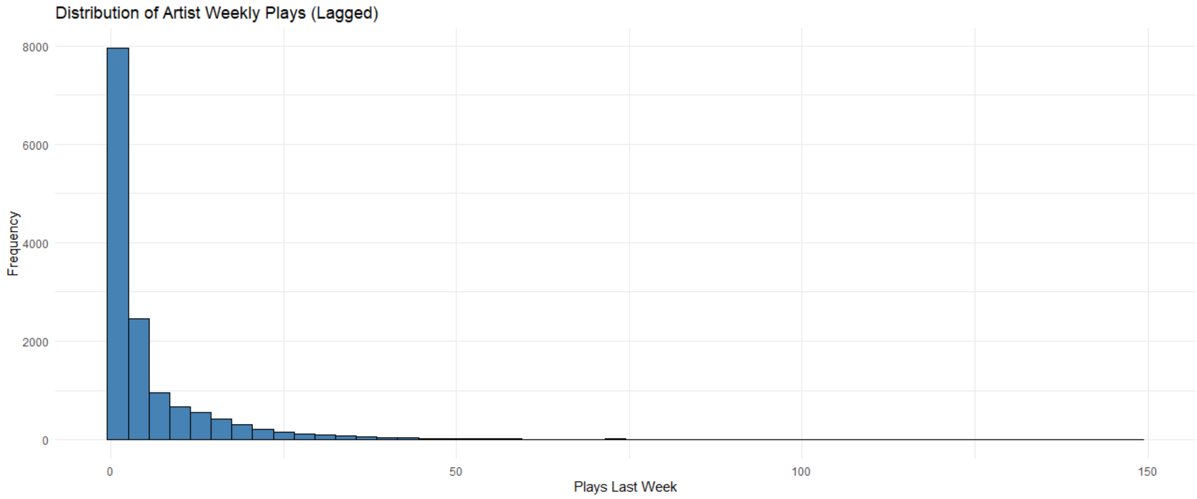
	Coefficient	P-value
log(FirstMonthPlays)	2.35036***	1.79e-10
log(FirstWeekPlays)	-0.51833	0.1639
log(BingeFactor)	7.01288***	1.22e-07
Growth Week2	0.03648*	0.0318
Consistency Month1	-1.87666**	0.0051
Constant	-11.22339	<2.2e-16

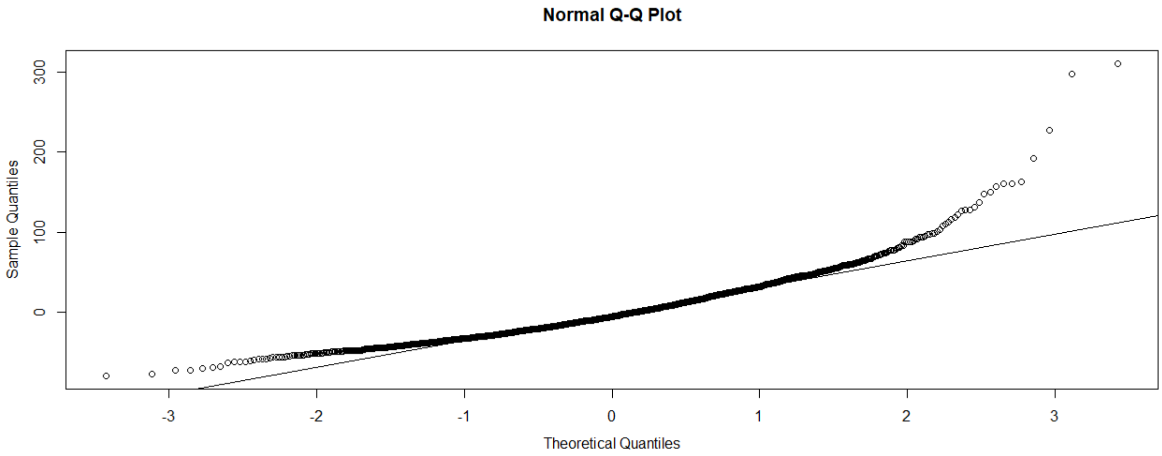
Confusion Matrix

	Actual	
Predicted	0	1
0	680	14
1	13	28

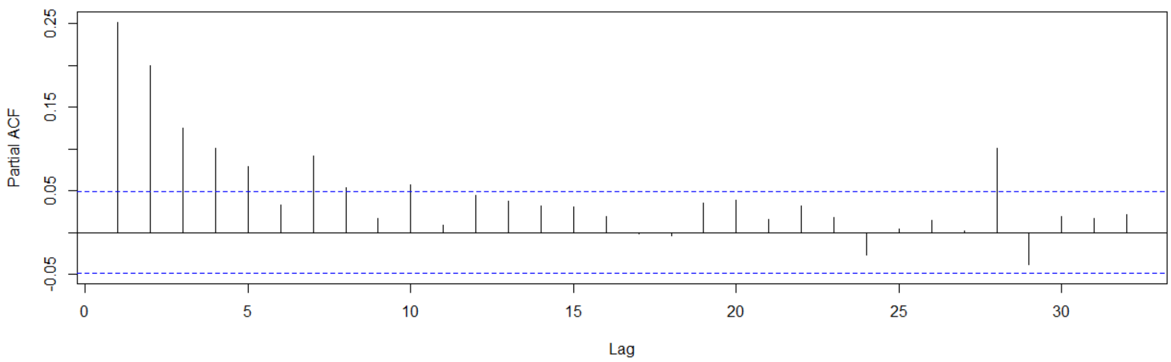
The visual suite



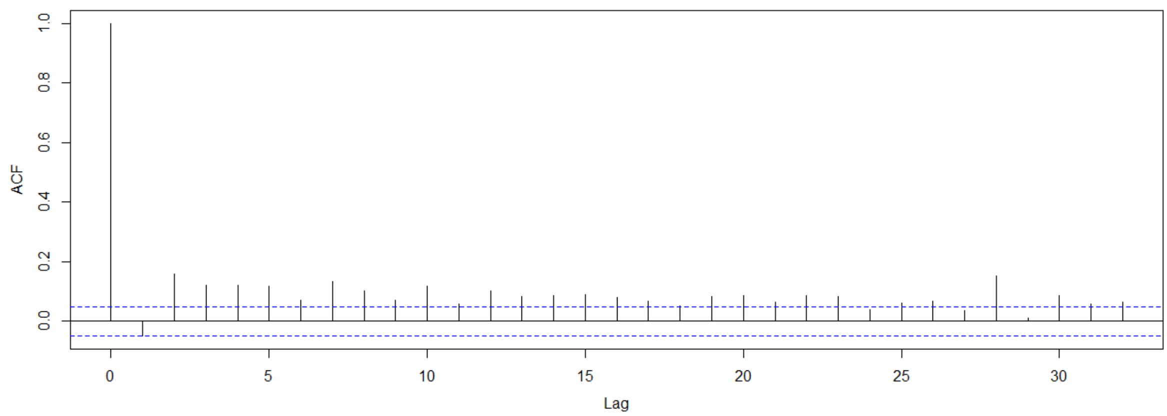




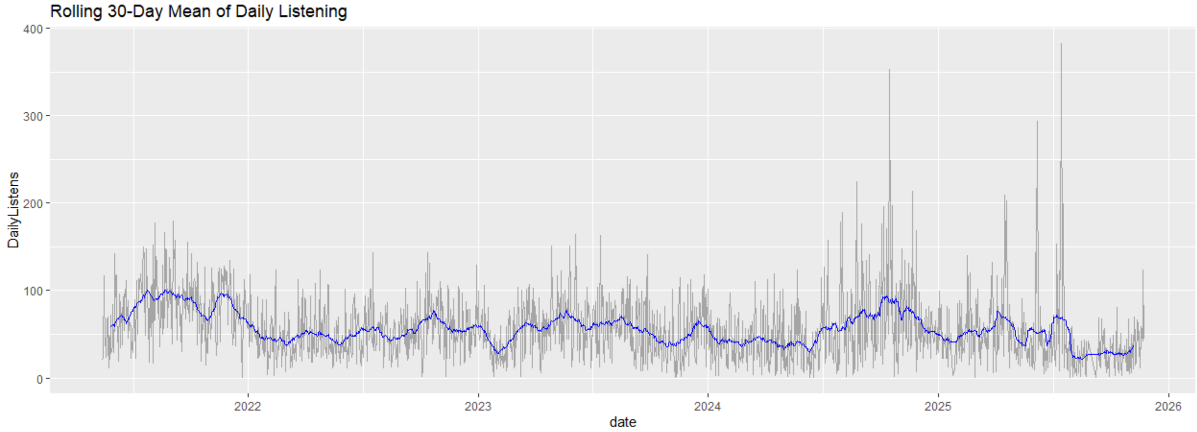
PACF of Daily Listening



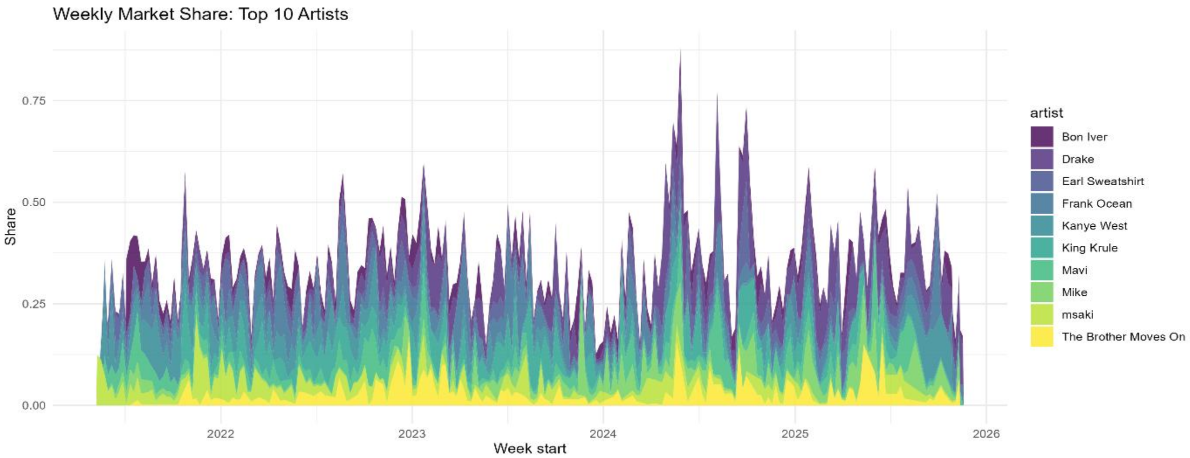
Residual ACF of AR(1) Model

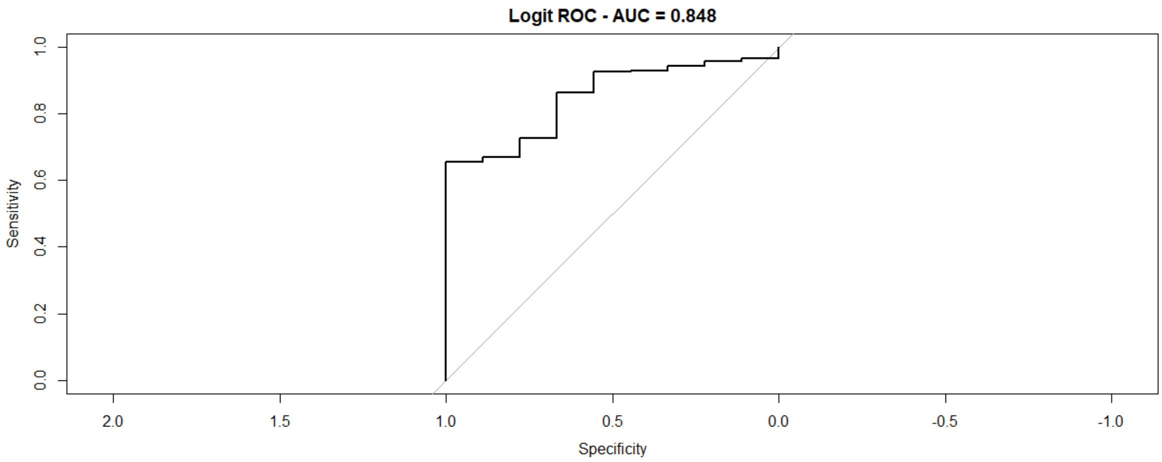


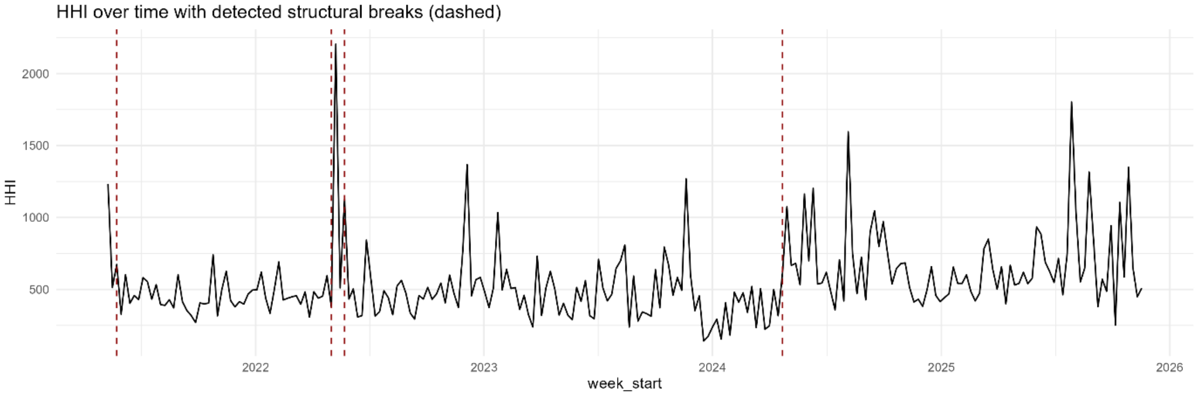
Rolling 30-Day Mean of Daily Listening




Rolling 30-Day Volatility of Daily Listening
Standard Deviation of Daily Scrobbles
80
60
40
20
2022
2023
2024
2025
2026
Date


Weekly Market Share: Top 10 Artists
Share
0.75
0.50
0.25
0.00
2022
2023
2024
2025
2026
Week start
artist
Bon Iver
Drake
Earl Sweatshirt
Frank Ocean
Kanye West
King Krule
Mavi
Mike
msaki
The Brother Moves On


Logit ROC - AUC = 0.848
Sensitivity
1.0
0.8
0.6
0.4
0.2
0.0
2.0
1.5
1.0
0.5
0.0
-0.5
-1.0
Specificity



Thank you for even getting this far. I do apologise for the unconventional presentation of the statistical results and the appendix as a whole.